PHY · Science inquiry skills — syllabus points and past-paper evidence

Every award of the 2016–2025 papers, labelled to the 2026/2027 Year 12 syllabus content (identical in both). Statistics from the labels; notes written by AI from that evidence: review before use.

Pattern 1: Questions, hypotheses and predictions

Syllabus: S26-SIS-SIS-01 / S27-SIS-SIS-01

Frequency: 0 of 10 exam years; 0 marks in all.

Summary: This point covers the front end of a scientific investigation: turning an observation into a testable question, stating a hypothesis, and predicting what should happen. The point has no past-paper parts recorded (parts_total 0), so this note is written from the syllabus wording and general physics practice, not from examiner awards. It is usually assessed inside investigation or design-based questions rather than as a stand-alone topic.

Recognise: Look for prompts that ask you to suggest what could be investigated, write a question an experiment could answer, state what you expect to happen and why, or say what a result would show. Typical wording includes 'identify a question', 'propose a hypothesis', 'predict the outcome', or 'what would you expect if'.

Steps

1. Start from the observation or the stated context and identify the single variable or relationship that is unknown.
2. Write a question that names the independent variable, the dependent variable, and the system, for example: does increasing the mass of a trolley change its acceleration down a fixed slope?
3. Form a hypothesis as a testable statement of cause and effect, linking the variables through a physical principle, for example: acceleration is proportional to net force, so doubling the force should double the acceleration.
4. Predict the outcome by applying the relevant relationship (for example $a = F/m$ or $v = u + at$) and state the direction and rough size of the change.
5. Say what result would support the hypothesis and what result would contradict it, so the prediction can be tested.
6. Justify the prediction briefly with the physics principle rather than only stating a guess.

Where the marks are

- A mark for a question that is focused, answerable by experiment, and names the variables.
- A mark for a hypothesis that is testable and states a cause-and-effect relationship.
- A mark for a prediction that is consistent with a named physics principle and gives the direction of change.
- A mark for linking the prediction to an observable result, such as a measurement that would confirm or refute it.

Pitfalls

- Writing a question that cannot be answered by measurement, such as one that asks for an opinion.
- Stating a hypothesis that is only a restatement of the observation, with no proposed relationship.
- Predicting a result without giving a reason from physics, so the prediction earns no justification mark.
- Confusing the independent and dependent variables when writing the question or hypothesis.
- Giving a prediction with the wrong direction, such as saying acceleration decreases when the force increases.

Pattern 2: Designing investigations

Syllabus: S26-SIS-SIS-02 / S27-SIS-SIS-02

Frequency: 1 of 10 exam years; 2 marks in all.

Marks reward: Investigation design 2

Past questions: 2018P-15(f)

Summary: This point asks you to plan an investigation end to end: the procedure, the materials, and which primary or secondary data to collect, along with safety and ethics. It usually appears as a question asking you to identify sources of experimental error in a practical and say how to make the result more accurate or reliable.

Recognise: Look for wording such as describe, suggest, or improve an experiment, sources of error, how to make the result more accurate, or what equipment and data would be needed. Often it follows a completed practical, so the answer should be grounded in the set-up already described.

Steps

1. Identify the independent variable you change, the dependent variable you measure, and the controlled variables you hold constant.
2. Write the procedure in ordered steps that a classmate could follow, including how many trials you take at each setting and how you record the readings.
3. List the materials and choose instruments whose range and resolution suit the quantity being measured.
4. Decide what data to collect: primary data from direct measurement, repeated trials to get a mean and spread, and any secondary data taken from a reliable source if it is needed.
5. For each error source, say whether it is random (scatter between readings) or systematic (a consistent offset such as a zero error or a miscalibrated instrument).
6. Link each modification to the error it reduces, for example repeating readings and averaging for random error, or calibrating or using a more precise instrument for systematic error.
7. Note any hazards and how to manage them, and any ethical issues such as handling living organisms or data from people.

Where the marks are

- A mark for naming a specific, realistic source of error that belongs to this set-up, not a generic one.
- A mark for a modification that would directly reduce that named error and would produce a more accurate result.
- Marks are given for each correct, distinct error and its matching improvement, so two clearly explained pairs can earn the full allocation.
- Vague answers such as human error or bad luck do not earn credit unless they are made specific.

Pitfalls

- Writing a source of error without saying how to modify the method to reduce it.
- Proposing a change that does not address the error named, such as buying a more expensive instrument when the problem is a parallax reading.
- Confusing random error with systematic error, so the fix does not match the problem.
- Listing many weak points instead of two or three well-explained, specific ones.
- Describing only the apparatus and forgetting to say what measurements are taken and how many.

Pattern 3: Conducting practical work

Syllabus: S26-SIS-SIS-03 / S27-SIS-SIS-03

Frequency: 0 of 10 exam years; 0 marks in all.

Summary: This point covers carrying out practical work safely, competently and in an orderly way, so that the data collected are valid and reliable. The source data records no past-paper parts for this point, so this note is written from the syllabus wording and general laboratory practice alone; in exams it is expected to appear inside experimental-design and data-collection questions rather than as a stand-alone item.

Recognise: A question that describes an apparatus set-up or a procedure and asks what should be done, what must be checked, what hazard exists, or why the results cannot be trusted. Key words include zeroing, calibration, range, precautions, controlling variables, repeat trials and anomalous results.

Steps

1. State the aim, the dependent variable being measured, the independent variable being changed, and the variables that must be held constant.
2. Choose an instrument whose range and smallest division suit the expected values, and estimate the precision of each reading.
3. Identify hazards first (electricity, heat, chemicals, glassware, moving parts), wear suitable protection, secure the set-up and use insulated leads.
4. Assemble the apparatus in a logical order, check connections, and zero or calibrate each instrument before any data are taken.
5. Change only the independent variable between settings, and record every reading with units and an uncertainty (for an analogue scale, about half the smallest division).
6. Repeat each setting, look for values that clearly disagree with the rest, and justify any exclusion; relative uncertainty = absolute uncertainty / measured value x 100%.
7. Shut down the supply and equipment in a safe order, then judge validity (does it measure what it should?) and reliability (are repeats consistent?).

Where the marks are

- A mark for naming a relevant hazard and a suitable precaution before the procedure starts.
- A mark for correct instrument use: zeroing or calibration, an appropriate range, and reading with the eye level with the scale.
- A mark for an ordered method that changes one variable at a time and controls the others.
- A mark for a clear data table with units and uncertainties on every recorded value.
- A mark for repeating trials and identifying or justifying the handling of anomalous results.

Pitfalls

- Reading an analogue scale from an angle, which introduces parallax error.
- Forgetting to zero or calibrate an instrument before use, so every reading is offset.
- Changing more than one variable at a time, so the cause of a change cannot be identified.
- Recording values without units, or with more significant figures than the instrument can resolve.
- Measuring only once, or discarding an anomalous value without a stated reason.

Pattern 4: Representing data: SI units, symbols, significant figures

Syllabus: S26-SIS-SIS-04 / S27-SIS-SIS-04

Frequency: 10 of 10 exam years; 64 marks in all.

Marks reward: Units, significant figures, direction 48, Graph work 10, Final answer 6

Past questions: 2016P-16(d); 2016R-13; 2017C-21(d); 2017C-22(b); 2017P-19(b); 2017P-19(c); 2017R-13(b); 2018P-15(a); 2018P-15(b); 2019P-17(a); 2019P-17(b); 2019P-17(c); 2020C-21(c); 2020P-19(a); 2020P-19(b); 2021C-20(ei); 2021C-20(eii); 2021P-13(a); 2021P-13(b); 2021P-13(c); 2021P-16(a); 2021P-16(b); 2021P-16(c); 2022C-18(e)...

Summary: This point tests whether a student can present data correctly: final answers and graph axes must carry the right SI units and symbols, and numerical answers must be rounded to an appropriate number of significant figures. It appears mostly as a single presentation mark attached to a calculated or estimated answer, and as axis and plotting marks in practical-based graph questions.

Recognise: Any question that ends with 'Express your answer in appropriate significant figures', 'Give your answer to two significant figures', 'give its units', or a blank labelled Answer with a unit after it. Also any graph task that asks you to label axes, plot data, or draw a line of best fit, and any estimate question where the marks depend on rounding.

Steps

1. Do the whole calculation with full precision on the calculator and only round at the end, so that rounding errors do not change the last significant figure.
2. Decide the number of significant figures from the least precise data given in the question, and follow any explicit instruction in the question; for estimates, usually one or two significant figures is enough.
3. Convert every quantity to SI base or derived units before calculating (for example grams to kilograms, centimetres to metres, millitesla to tesla), so the result comes out in SI units automatically.
4. Write the unit as its correct SI symbol with the right case and spacing, such as m s-1, V, T, Hz, N or J, and make sure the unit matches the physical quantity (speed, potential difference, energy, and so on).
5. For a graph, label each axis with the quantity name and its SI unit symbol, choose a scale that spreads the points over most of the grid, plot points accurately, and draw one straight line of best fit through the trend, not necessarily through the origin if the data does not pass through it.
6. For a gradient or intercept read from a graph, show the construction triangle, take coordinates with their units, and give the derived unit as the y-unit divided by the x-unit, rounded to the same significant figures as the data.

Where the marks are

- One mark is usually given for the number rounded to the required significant figures, separate from the mark for the calculation itself.
- A separate mark is often given for the correct unit attached to the answer, even when the numerical value is right.
- Graph marks are commonly split into axis orientation, axis labels including units, and accurate plotting of each point.
- Gradients and intercepts read from a graph must carry a unit derived from the axis units, and the value must be stated to suitable significant figures.
- Estimates still earn the presentation mark when the value is given to a sensible number of significant figures, so a rough answer should be rounded, not left with many digits.

Pitfalls

- Copying all the digits shown by the calculator instead of rounding to the required significant figures, which loses the presentation mark even when the value is correct.
- Leaving the unit off, writing a non-SI unit such as km/h or cm when the answer should be in SI, or using an incorrect symbol or case (for example M for m, or t for T).
- Forgetting to convert prefixed units such as mT, cm or g before calculating, so the number and unit are both wrong.
- Labelling a graph axis with the quantity but no unit, or writing units in a place or form that does not match the axis.
- Rounding intermediate values too early, which can change the final significant figures, or giving a gradient unit that does not match the ratio of the two axis units.

Pattern 5: Trends, patterns and relationships in data

Syllabus: S26-SIS-SIS-05 / S27-SIS-SIS-05

Frequency: 5 of 10 exam years; 6 marks in all.

Marks reward: Graph work 3, Final answer 2, Substitution and working 1

Past questions: 2016C-22(d); 2020P-13(e); 2021P-16(a); 2022C-18(b); 2023P-16(b)

Summary: This point examines whether students can take raw or tabulated measurements, process them (calculate derived quantities, transform variables, plot them) and read a trend, pattern or relationship from the result. In the papers it appears as short parts that ask for a calculated column, a plotted or read-off value, or a one-line conclusion about what the data show, often tied to a practical or a given table.

Recognise: Look for a table of measurements with a column to complete, a request to plot or use a graph, or

a prompt such as 'what does the data suggest', 'identify the relationship' or 'estimate from the trend'. Watch for a need to transform a variable (squaring, reciprocal, dividing by a quantity) so that a straight-line relationship can be seen, and for questions asking which value the data best support.

Steps

1. Check the units and the required significant figures or notation before calculating anything.
2. Calculate each derived column with the correct operation (for example square a frequency, or divide by a reference quantity), keeping extra digits until the end.
3. Plot the transformed variables with suitable axes, scales and labels, and draw a line or curve of best fit through the points.
4. Identify the form of the relationship: linear if the points fall on a straight line through the origin (y = kx), proportional if y/x is constant, inverse if y x is constant, or otherwise a curve.
5. Find the gradient from two well-separated points on the line of best fit, and relate it to the physical constant it represents, using the correct units.
6. Read values off the graph or table to support a conclusion, and state the trend in a sentence that uses the variables and direction of change.

Where the marks are

- Marks are given for a correct calculated or transformed value taken from the table or the graph, with the values used shown.
- A mark is commonly given for a plotted or read-off value that is consistent with the data, rather than only for a final number.
- A mark is commonly given for stating the pattern clearly, such as the variables increasing together or one falling as the other rises.
- Marks are given for giving the answer to the required significant figures and in the required form, such as scientific notation.
- Where a value is estimated from data, the examiner looks for the single best answer the data support, not a value that merely fits one point.

Pitfalls

- Squaring or otherwise transforming the wrong column, or taking the square root instead of the square when linearising.
- Losing the mark for significant figures or for expressing a very small or large value in scientific notation.
- Stating a trend without the direction of change or without naming both variables.
- Drawing a line that does not pass sensibly through the points, or calculating a gradient from two points that are too close together.
- Reading a value off a graph using the wrong axis, or mixing up which variable is the dependent variable.

Pattern 6: Random and systematic uncertainty

Syllabus: S26-SIS-SIS-06 / S27-SIS-SIS-06

Frequency: 4 of 10 exam years; 7 marks in all.

Marks reward: Uncertainty and error 7

Past questions: 2016R-13；2018P-15(f)；2019C-20(aii)；2023P-16(e)

Summary: This point asks you to tell random uncertainty (unpredictable scatter between readings) apart from systematic uncertainty (a consistent bias in the measurement or the model), and to estimate how uncertainty in the inputs affects a final result. In the papers it usually appears as a short written item after a calculation: name a source of error, explain why an answer is only an estimate, or describe an assumption that may affect accuracy.

Recognise: Look for wording such as 'identify a source of error', 'give a reason why your answer is only an

estimate', 'describe an assumption that may affect accuracy', or 'how could the result be made more accurate'. A calculation where one input is given as a central value plus or minus a range, followed by a question about how reliable the answer is, is also this point.

Steps

1. Write down the formula used and list each input with its uncertainty. If an input is an estimate, such as a centre of mass position assumed to lie within a range, give the range explicitly.
2. Carry the uncertainty through the calculation: recompute the result at the low and high ends of the input range to see how far the answer can move, or state the direction in which the answer shifts.
3. Classify each source. Random: scatter from reading an instrument by eye, reaction time, or small fluctuations, which shows up as differences between repeat readings. Systematic: a fixed offset from a miscalibrated or zero-error instrument, or a simplifying assumption such as treating a quantity as constant when it actually changes.
4. For a 'why is this only an estimate' question, name the specific input or assumption that is approximate and say what physical effect it ignores or simplifies.
5. For a 'how to improve' question, match the fix to the type: repeat trials and average to reduce random scatter; calibrate, use a better instrument, or change the method or model to remove systematic bias.
6. End with a clear conclusion: the result is an estimate, and its reliability depends on the named uncertainty, usually stated as a range.

Where the marks are

- One mark for each specific, valid source of uncertainty that is linked to this experiment, not a generic phrase.
- One mark for identifying an assumption and explaining why it is not exactly true in the situation (the physical mechanism, not just 'it changes').
- One mark for using the given uncertainty range in the calculation and carrying it through to a stated effect on the answer.
- One mark for stating the direction or size of the effect, not only the name of the source.
- For improvement questions, one mark for a fix that directly targets the named source.

Pitfalls

- Writing generic answers such as 'human error' or 'inaccurate equipment' when the question excludes them or does not reward them.
- Mixing up random and systematic: repeating a measurement reduces random scatter but does not remove a systematic offset.
- Naming a source without saying how it affects the result.
- Describing an assumption vaguely (for example 'the wire changes') without saying which quantity changes and why it matters.
- Giving an uncertainty range but not using it to show how the final answer is affected.

Pattern 7: Absolute and percentage uncertainty

Syllabus: S26-SIS-SIS-07 / S27-SIS-SIS-07

Frequency: 4 of 10 exam years; 9 marks in all.

Marks reward: Uncertainty and error 7, Final answer 1, Units, significant figures, direction 1

Past questions: 2017R-13(b)；2019P-17(d)；2021P-16(e)；2021P-16(f)；2024R-10

Summary: This point checks that you can state the absolute uncertainty of a measured value and turn it into a percentage uncertainty, then judge whether a result is precise enough. In the papers it appears mostly as a short calculation or comment attached to a graph, a density or a trigonometric value, and it is usually worth a few marks rather than a whole question.

Recognise: Look for a measured value with a tolerance written as ±, a ruler or instrument reading, a gradient or result read from error bars, or a prompt to 'include the uncertainty', 'express as a percentage' or

'comment on the uncertainty'. A question asking how the percentage error changes as the measured size changes is also testing this point.

Steps

1. Write the measured value and its absolute uncertainty with the same units, for example x ± Δx.
2. Calculate the percentage uncertainty as (Δx / x) × 100%, using the measured value x, not the uncertainty.
3. If the result is a product or quotient of measured values, add their percentage uncertainties to estimate the percentage uncertainty of the result.
4. If the result is a power, multiply the percentage uncertainty by the power; for example, a cube has three times the percentage uncertainty of its side length.
5. Convert back to an absolute uncertainty by multiplying the result by the percentage uncertainty, and write the final answer as value ± absolute uncertainty with appropriate significant figures.
6. For a function such as a trigonometric ratio, find the change in the output when the input changes by its uncertainty, then divide by the output to get the percentage error.
7. Comment on the uncertainty by stating whether the percentage is small enough for the stated precision or how it changes with the size of the measurement.

Where the marks are

- Marks are given for the correct percentage calculation, with the uncertainty divided by the measured value rather than by the uncertainty itself.
- A separate mark is often given for expressing the answer as a percentage or as value ± absolute uncertainty with the right units.
- Marks are given for stating the trend or conclusion clearly, such as whether the percentage error rises or falls as the measurement changes, supported by at least two calculated values.
- A mark is given for correctly using the input value and its error when a function such as a trigonometric ratio is involved.
- Marks are given for a correct final answer with suitable significant figures and, where asked, the absolute uncertainty included.

Pitfalls

- Dividing the absolute uncertainty by the wrong value, or by the uncertainty itself, so the percentage is wrong from the start.
- Forgetting to multiply by 100 to convert the fraction into a percentage, or giving a percentage without the % sign.
- Using the total range of a reading rather than the stated ± tolerance, or ignoring the given tolerance on an instrument.
- Failing to raise the percentage uncertainty to the power in a formula, such as the cube of a length, so the density uncertainty is underestimated.
- Giving the answer to too many significant figures, or dropping the ± uncertainty when the question asks to include it.

Pattern 8: Combining uncertainties

Syllabus: S26-SIS-SIS-08 / S27-SIS-SIS-08

Frequency: 4 of 10 exam years; 13 marks in all.

Marks reward: Uncertainty and error 10, Evaluate and conclude 1, Substitution and working 1, Final answer 1

Past questions: 2017R-13(b)；2019P-17(d)；2021P-16(f)；2024R-10

Summary: This point asks you to work out the overall uncertainty of a calculated result from the uncertainties of the measured values that go into it. It usually appears as a multi-part question where a derived quantity (a graph gradient, a volume, a density, or a trigonometric value) must be given as value plus or minus uncertainty, sometimes followed by a judgement on whether the uncertainty is acceptable.

Recognise: Any question that gives measurements written as value ± uncertainty, or asks for the uncertainty or percentage error of a result built from two or more measured values. Also look for a gradient read from a graph with error bars, or a result that involves powers, division or a trigonometric function.

Steps

1. Write each measured quantity with its absolute uncertainty, and where needed its relative (percentage) uncertainty = absolute uncertainty ÷ measured value × 100%.
2. For addition or subtraction, add the absolute uncertainties of the terms.
3. For multiplication or division, add the percentage uncertainties, then convert back to an absolute uncertainty by multiplying by the calculated value.
4. For a power, multiply the percentage uncertainty of the base by the power (e.g. for volume $V = L^3$, the relative uncertainty of V is 3 times the relative uncertainty of L).
5. For a gradient from a graph, choose two points far apart on the line of best fit, find Δy and Δx with their uncertainties from the error bars, and combine them; alternatively take the steepest and shallowest lines through the error bars and halve the difference between their gradients.
6. For a non-linear function such as sine, substitute the value plus and minus its uncertainty to find the range of the output, or estimate the uncertainty as $|f'(x)| \times \Delta x$, then write the result as value ± uncertainty.

Where the marks are

- A mark for stating or using the correct absolute or percentage uncertainty for each measured value before combining them.
- A mark for choosing the correct rule for the operation: absolute uncertainties add for addition and subtraction, percentage uncertainties add for multiplication and division.
- A mark for the correct combined uncertainty, presented as a ± value alongside the answer with matching units and sensible significant figures.
- For gradient questions, a mark for naming the two points used and showing their Δ values with uncertainties.
- For comment or evaluation questions, a mark for comparing the percentage uncertainty against the stated acceptable range and giving a clear conclusion.

Pitfalls

- Adding percentage uncertainties when the operation is addition or subtraction, or adding absolute uncertainties when the operation is multiplication or division.
- Forgetting to convert a combined percentage uncertainty back to an absolute value at the end.
- Forgetting the power factor when a quantity is squared or cubed, such as for area or volume.
- Using the wrong measured value as the base for the percentage, or mixing units such as mm and cm within one calculation.
- Giving the uncertainty to too many significant figures, or not rounding the answer to match the precision of its uncertainty.

Pattern 9: Anomalous data and percentage difference

Syllabus: S26-SIS-SIS-09 / S27-SIS-SIS-09

Frequency: 2 of 10 exam years; 4 marks in all.

Marks reward: Uncertainty and error 4

Past questions: 2019P-17(a)； 2025P-15(c)

Summary: This point tests whether students can spot a data point that clearly departs from the trend in experimental results and compare a measured value with an accepted value using percentage difference. In the sampled papers it appears as a graphing part that asks for an outlier to be identified, and as a short calculation part that asks for a percentage difference against a given accepted value.

Recognise: Look for wording such as 'anomalous', 'outlier', 'does not fit the trend', 'compare with the

accepted value', or 'percentage difference'. Typically it follows a graph or a set of measured results, or sits after a calculation that produces an experimental value to be checked against a reference value.

Steps

1. Plot or inspect the data and identify the point that lies well away from the trend or line of best fit; name it clearly.
2. Justify it as anomalous (for example a likely measurement or recording error) and state whether it is excluded from the line of best fit.
3. Identify the experimental value and the accepted value, and make sure both are in the same units, taking care with scientific notation.
4. Write the numerator as the difference between the experimental value and the accepted value.
5. Use the accepted value as the denominator: percentage difference = (experimental minus accepted) / accepted x 100%.
6. Give the result with the % sign and at a sensible number of significant figures.

Where the marks are

- One mark for clearly identifying the anomalous point, usually on the graph or in a written statement.
- One mark for writing the numerator as the difference between the two values, with correct units and order of magnitude.
- One mark for using the accepted (reference) value as the denominator.
- One mark for expressing the final result as a percentage with the % sign.

Pitfalls

- Removing a point without a reason, or calling a point an outlier when it actually lies on the trend.
- Using the experimental value as the denominator instead of the accepted value.
- Forgetting to multiply by 100 or to write the % sign.
- Losing units or powers of ten when subtracting values written in scientific notation.
- Giving only the final percentage without showing the difference and the denominator used.

Pattern 10: Evidence-based conclusions

Syllabus: S26-SIS-SIS-10 / S27-SIS-SIS-10

Frequency: 4 of 10 exam years; 10 marks in all.

Marks reward: Evaluate and conclude 10

Past questions: 2016C-22(b); 2016C-22(d); 2016C-22(e); 2018C-18(a); 2018R-6(b); 2019P-17(d); 2023C-19(d)

Summary: This point asks you to choose the relevant evidence, combine it, and draw a conclusion you can defend. In the papers it almost always appears as the final part of a context question, asking whether a result supports a claim, which option is more likely, or whether a rule was obeyed, and it is marked as EVALUATE rather than as a standalone skill.

Recognise: Look for words such as justify, explain whether, is there evidence, which is more likely, or is it consistent with. The question gives data, a graph, a spectrum, a table, a calculation or an experiment and asks what it shows. Any question that ends with a yes or no and then asks for a reason is this point.

Steps

1. Pick out the piece of data or observation that bears on the question, and say in one line what it shows (a peak position, a measured value, a calculated total, a shift).
2. Compare it with a reference: a table, a known value, a prediction from theory, or a stated threshold. Ask whether it matches or falls outside that reference.
3. Link the variable to the conclusion with the right relationship, for example photon energy $E = hc/\lambda$ to turn a wavelength into an energy, or the sum of percentage uncertainties to test whether a gradient is accurate enough, or total charge before = total charge after.

4. If the question asks which option is more likely, state the option and the reason it is favoured, such as a higher intensity or a more common process.
5. Make a clear claim (yes, no, consistent, forgery, wave-like, more likely) and attach the evidence in the same sentence, so the claim and its reason are read together.
6. Check for a limit: a missing value, an anomaly or a precision problem can weaken the claim, and a sound answer says so briefly.

Where the marks are

- One mark for a conclusion that is clearly stated and matches the evidence (yes or no, which one, or what the data means).
- One mark for pointing to the specific value, peak or calculation that supports the conclusion, rather than a general statement.
- One mark for a comparison against a reference value or range, including a check that the difference lies within an uncertainty.
- Where a question asks for a chain of reasoning, each linked step (e.g. an event that could not have happened yet, followed by what that implies) earns its own mark.
- For a calculation-based justification, the final percentage or total must be stated and tied back to the yes or no answer.

Pitfalls

- Stating a conclusion but not naming the piece of evidence that supports it, so the justification mark is lost.
- Quoting a value without comparing it to the reference, leaving the examiner to infer the match.
- Calling a difference acceptable without checking it against the uncertainty or the stated tolerance.
- Answering with a vague word such as likely or probably without a physical reason.
- Ignoring the second half of a question that asks which option is more likely, or giving only one side of the comparison.

Pattern 11: Interpreting texts and evaluating claims

Syllabus: S26-SIS-SIS-11 / S27-SIS-SIS-11

Frequency: 0 of 10 exam years; 0 marks in all.

Summary: This point covers reading a scientific text (an article, data summary, or experiment write-up), judging whether its method and conclusion are supported by the evidence, and building a reasoned argument of your own. It is an inquiry-skills point, so it usually appears as a stimulus passage or data set followed by questions asking you to comment on the process or claim, not as a stand-alone content topic. Note: the local file records no past-paper parts or marks for this point (parts_total 0), so this note is written from the syllabus wording and general physics practice only.

Recognise: Look for a short passage or results table followed by verbs such as evaluate, comment on, assess the validity of, justify, or 'do the data support the claim'. Questions may give a claim from a news item or a student, or ask you to identify a weakness in a method and then suggest how to fix it. Often two or three parts build on each other: first find the evidence, then judge it, then argue a conclusion.

Steps

1. Read the text once for the aim, the independent and dependent variables, and what is claimed.
2. List the evidence given: measured values, units, number of trials, and any controls.
3. Check the process: were variables controlled, is the measurement precise and accurate, is the sample size adequate, and are there any uncontrolled factors.
4. Test the conclusion against the evidence: does the data show the claimed relationship, is the trend strong enough, and could another explanation fit equally well.
5. Identify limitations and state their effect on the conclusion (for example, a small sample means a weak claim, not a false one).

6. Build the argument as claim, then supporting evidence, then reasoning linking evidence to claim, then a qualification or suggested improvement.
7. Keep the judgement proportional: say 'the data supports' or 'only partly supports' where the evidence allows, not 'proves'.

Where the marks are

- Marks usually go for correctly locating the relevant evidence in the text or table before judging it.
- Marks go for naming a specific, physics-based flaw (uncontrolled variable, too few trials, unit or scale problem) rather than a vague one such as 'human error'.
- Marks go for linking each judgement to the evidence with reasoning, not just stating an opinion.
- Marks go for a clear, justified overall verdict on whether the conclusion follows from the data.
- Marks go for suggesting a realistic improvement that addresses the specific weakness identified.

Pitfalls

- Saying the experiment is 'inaccurate' or 'not good' without naming what made it so.
- Calling a result 'proven' from a small or single set of data.
- Ignoring units, scale, or the range of the data when judging a trend.
- Suggesting an improvement that does not address the flaw you identified, such as asking for more time when the issue is an uncontrolled variable.
- Restating the claim as the answer without weighing the evidence for and against it.

Pattern 12: Representations that communicate and predict

Syllabus: S26-SIS-SIS-12 / S27-SIS-SIS-12

Frequency: 5 of 10 exam years; 13 marks in all.

Marks reward: Graph work 13

Past questions: 2016C-21(e); 2018P-15(b); 2019P-17(a); 2022C-18(b); 2025C-20(d)

Summary: This point tests whether you can choose, draw, or modify a graph (or a clear text description of one) to show a physical relationship, then use it to communicate a result or predict behaviour. In the papers it appears as sketch, plot, or draw questions: either plotting measured data with a line of best fit, or altering a given graph to show how a stated change affects the curve.

Recognise: Look for command words such as plot, draw, sketch, graph, or modify, often with a grid, axes, or an existing curve supplied. Phrases like 'on the same graph', 'include the line of best fit', or 'reference the graphs' signal this point. A question that asks you to explain a claim by pointing to a graph is also testing representation.

Steps

1. Read the axes first: identify the dependent and independent quantities, their units, and what the question wants on each axis. If the relationship is curved, consider whether plotting against a transformed variable (such as a reciprocal or a square) would make it straight.
2. Choose a scale that spreads the data across most of the grid, uses sensible round steps, and starts the axes at zero only when that is meaningful.
3. Plot each point accurately and label both axes with the quantity and unit.
4. Draw one smooth line or straight line of best fit through the trend, balancing points above and below it. Do not force it through the origin unless the theory requires it, and do not join the points dot-to-dot.
5. For a modified graph, start from the given curve and decide which features the change affects, such as a shift, a change in height, a new minimum, or a change in overall intensity. Redraw only those features and keep the features that should not change in exactly the same positions.
6. Check the sketch against the physics: the curve should return to its baseline where it should, peak positions should remain where the stated conditions say, and the graph should not cross a reference line the question says to avoid.

Where the marks are

- Marks are given for correctly labelled axes with units, and for an appropriate, evenly spaced scale that uses the grid.
- Accurate plotting of the points and a single straight line of best fit that follows the trend and is not forced through the origin when it should not be.
- For sketches of a modified graph, each feature that changes correctly earns a mark, such as the start of a dip shifting, the curve dropping below a reference, or a flat base being kept.
- Features that should stay fixed, such as characteristic peak positions, are rewarded when they are kept in place.
- Where a written answer must refer to graphs, a mark is given for stating the specific region or feature of the graph that supports the claim.

Pitfalls

- Joining points dot-to-dot instead of drawing a line of best fit, or drawing a line through the origin when the data does not go there.
- Missing axis labels or units, or using an awkward scale that cramps the data into a corner of the grid.
- Sketching a modified curve that crosses the reference line, fails to return to baseline, or moves peaks that should stay fixed.
- Drawing on the wrong sheet or ignoring instructions about cross-outs and spare grids, so the marker cannot see the attempt.
- Writing a justification that does not point to the relevant part of the graph, so the claim earns no mark even when the idea is right.

Pattern 13: Linear and non-linear graphs and models

Syllabus: S26-SIS-SIS-13 / S27-SIS-SIS-13

Frequency: 7 of 10 exam years; 46 marks in all.

Marks reward: Graph work 44, Substitution and working 1, Select the principle or relationship 1

Past questions: 2016P-20(b); 2019R-10(a); 2019R-10(b); 2020P-17(bi); 2020P-19(a); 2020P-19(cii); 2020R-7(b); 2022P-15(b); 2022R-8(b); 2023P-16(c); 2023R-8; 2024P-16(b); 2024P-17(b); 2024P-17(c); 2024R-2(b); 2025C-20(c); 2025C-20(e); 2025P-16(a); 2025P-16(b); 2025P-16(c); 2025P-16(e); 2025R-2

Summary: This point tests whether students can plot data or a physical law as a graph, read values from a graph, and rearrange a non-linear relationship so that it plots as a straight line for solving problems and predicting values. In the papers it appears mainly as graphing and graph-reading tasks, often in photoelectric, spectra, gravitation or electromagnetic-induction contexts, and a single question can carry several marks.

Recognise: Look for command words such as plot, graph, draw a line of best fit, read from the graph, estimate, or reorganise the data. The question usually supplies a table or a curve and asks you to add points, read off a value, or describe how to change the variables so the graph becomes a straight line.

Steps

1. Identify the independent variable (horizontal axis) and the dependent variable (vertical axis), and write the physical relationship as an equation, such as y = mx + c, or recognise a power law such as F proportional to 1/r squared.
2. If the relationship is non-linear, transform it so one quantity is plotted against a function of the other: for y proportional to 1/x, plot y against 1/x; for y = k x^n, plot y against x^n, or use log-log axes if the question allows.
3. Choose scales that spread the data over at least half the grid, and label each axis with the quantity and its unit.
4. Plot each point accurately, then draw one straight line of best fit (or smooth curve for non-linear data) that passes close to most points with roughly equal numbers on each side. Do not force it through the origin unless the physics requires it.

5. To read a value, find the given point on one axis, project a line to the graph and then across or down to the other axis. To find a gradient, pick two widely separated points on the line and calculate m = change in y over change in x.
6. Use the gradient and intercept to find an unknown constant, or substitute a value read from the graph into the relationship to make a prediction, showing the substitution.

Where the marks are

- A mark for each correctly plotted data point, usually allowed a small tolerance.
- A mark for axes that are labelled with quantities and units and have sensible, uniform scales.
- A mark for a straight line of best fit when the data are linear, or a smooth curve for non-linear data, not joined dot to dot.
- Marks for reading values with the working shown on the graph, such as projection lines, so the method is visible.
- A mark for correctly stating how to reorganise the data, such as plotting against 1/wavelength or stating both variables explicitly.

Pitfalls

- Plotting against the wrong variable when linearising, or describing a reorganised graph without naming both variables.
- Forcing the line through the origin when the data does not require it, or joining the points with straight segments.
- Missing axis labels or units, or choosing scales that squash all the data into one corner of the grid.
- Reading a value without drawing the projection lines on the graph, so the examiner cannot see how it was obtained.
- Calling the relationship directly or inversely proportional before describing the shape of the curve; describe the shape first, then name the relationship.

Pattern 14: Gradients and intercepts as physical quantities

Syllabus: S26-SIS-SIS-14 / S27-SIS-SIS-14

Frequency: 10 of 10 exam years; 49 marks in all.

Marks reward: Graph work 33, Final answer 5, Select the principle or relationship 5, Substitution and working 3, Qualitative explanation 3

Past questions: 2016P-20(c); 2017P-18(bii); 2017P-19(c); 2017R-13(a); 2018P-15(c); 2018P-15(d); 2018P-15(e); 2019P-17(b); 2019P-17(c); 2019R-10(c); 2020P-17(bii); 2020P-19(b); 2021P-16(c); 2022P-15(c); 2022P-15(d); 2023P-16(d); 2024P-16(c); 2024P-16(d); 2024P-16(e); 2025P-16(d)

Summary: This point asks you to read a straight-line graph, find its gradient or an axis intercept, and state what that number means physically, usually by matching it to a rearranged physics equation. In the papers it appears as a graphing task followed by a short follow-on asking what the gradient or intercept represents.

Recognise: A question says 'using your graph' to find a constant, or asks what the gradient or intercept represents. Look for a plot whose axes have been chosen (often by rearranging an equation) so the relationship is linear, such as plotting one quantity against the reciprocal of another.

Steps

1. Write the physics equation and rearrange it into y = mx + c, deciding which quantity sits on each axis.
2. Set the gradient equal to the combination of constants or quantities the rearrangement produces, for example a constant divided by another constant, or a constant multiplied by a numerical factor.
3. Draw the line of best fit, then choose two points on the line (not original data points), ideally well separated, and mark them clearly on the graph.
4. Calculate the gradient as rise over run, delta y divided by delta x, using the axis units and checking the sign.
5. Read where the line crosses an axis and interpret the intercept physically, for example as a threshold

value or as the value of the other quantity when the plotted variable is zero.

6. Solve for the unknown constant by equating the measured gradient to the expression from step 2, then give the answer with appropriate significant figures and units.

Where the marks are

- A mark for showing construction lines or clearly marking the two points used on the line of best fit.
- A mark for a correct gradient calculation with units and a sensible number of significant figures.
- A mark for stating that the gradient equals a named constant or ratio, not just giving a number.
- A mark for reading an intercept and interpreting it physically, such as a threshold value.
- A mark for linking the gradient to the rearranged equation before substituting values.

Pitfalls

- Using two original data points instead of two points on the line of best fit.
- Forgetting the gradient's unit, or using the wrong unit because the axes were transformed.
- Swapping the axes in the rise-over-run calculation, which inverts the physical constant.
- Equating the gradient directly to the constant and dropping a factor from the rearrangement, such as 2 pi or a charge term.
- Reading an intercept but not saying what it means physically.

Pattern 15: Dimensional analysis and units

Syllabus: S26-SIS-SIS-15 / S27-SIS-SIS-15

Frequency: 1 of 10 exam years; 1 marks in all.

Marks reward: Units, significant figures, direction 1

Past questions: 2024P-16(c)

Summary: This point checks that you can work out the correct unit of a calculated quantity, most often the gradient of a line of best fit, by tracking the units of the two axes. It appears in practical and data-analysis parts of the papers, usually as a short final step after a gradient calculation, and it is typically worth a single mark.

Recognise: Look for a question that asks you to calculate a gradient, rate, or other derived quantity from a graph you have drawn or been given, and to 'include the unit'. Any request to state the unit of a result after a calculation, or to check that an answer is dimensionally consistent, falls under this point.

Steps

1. Write down the unit of the y-axis and the unit of the x-axis, including any prefixes such as milli or kilo.
2. Write the gradient as a ratio: gradient = change in y / change in x, so its unit is (unit of y) divided by (unit of x).
3. Simplify the unit by cancelling and combining powers, for example writing a per-second quantity as s^-1 and a division of units as a single compound unit.
4. Check the unit against the physics: if the gradient represents a known quantity such as velocity, acceleration, or a force-to-mass ratio, confirm the unit matches the standard unit for that quantity.
5. Keep the unit alongside the numerical value to the required significant figures, and do not drop it when rounding.

Where the marks are

- One mark is generally given for stating the correct unit of the gradient, written in a recognised form such as a compound unit or with negative powers.
- The unit is credited when it follows from the axes you chose, so a correctly derived unit is accepted even if its written form differs from the standard one.
- The mark is separate from the numerical value and the significant figures, so both must be correct to earn the full credit where they are assessed together.

Pitfalls

- Quoting only the axis units (for example, writing the unit of y) and forgetting to divide by the unit of x.
- Writing the unit as a plain division without simplifying, or mixing up s and s^-1 when converting a per-second quantity.
- Ignoring prefixes on the axes, so a gradient in milli-units is given with the wrong scale.
- Stating a unit that does not match the quantity the gradient represents, which makes the answer dimensionally inconsistent.
- Omitting the unit entirely because the numerical part of the answer looks complete.

Pattern 16: Error bars and lines of best fit

Syllabus: S26-SIS-SIS-16 / S27-SIS-SIS-16

Frequency: 7 of 10 exam years; 10 marks in all.

Marks reward: Graph work 7, Uncertainty and error 3

Past questions: 2017P-19(b); 2017R-13(b); 2020P-19(a); 2021P-16(b); 2022P-15(b); 2023P-16(c); 2024P-16(b); 2025P-16(c)

Summary: This point examines whether students can show measurement uncertainty on a graph with error bars and draw a sensible line of best fit through the data. In the papers it appears mostly as a graphing task (plot the data, draw a straight line of best fit, with the line not forced through the origin unless the physics requires it), and occasionally as a follow-up where the gradient of that line is used to find its uncertainty from the maximum and minimum gradients.

Recognise: A graphing question that gives a table of measured values and asks you to plot them and draw a line of best fit, or a question that asks you to use the gradient and the error bars to comment on the uncertainty of a calculated value. Watch for wording such as 'include a line of best fit', 'draw error bars', 'comment on the uncertainty', or 'uncertainty in the gradient'.

Steps

1. Label both axes with quantity and unit, choose a scale that spreads the data over most of the grid, and plot each point accurately.
2. If uncertainties are given, draw a vertical and/or horizontal error bar through each point whose length is the stated uncertainty on each side of the point.
3. Draw one straight line that passes close to the middle of the points, with roughly as many points above as below, and that falls within the error bars where possible. Do not join the points dot-to-dot.
4. Only force the line through the origin if the physical relationship requires it; otherwise let the data decide the intercept.
5. To find the uncertainty in the gradient, choose two other points on the scatter (not the first two you used) to give the steepest possible line and the shallowest possible line that still pass through the error bars. Calculate each gradient as rise over run.
6. Take half the difference between the maximum and minimum gradients as the gradient uncertainty.
7. If the gradient is multiplied by a constant to give the quantity asked for (for example 2π), multiply the gradient uncertainty by that same constant and express the answer to the appropriate significant figures.

Where the marks are

- Marks are given for a clear, straight, accurate line of best fit that a marker can judge as a good fit.
- Marks are given for correctly labelled axes and points plotted accurately against the grid.
- Marks are given for choosing two different points to build each of the maximum and minimum gradients, and for showing each gradient calculation.
- Marks are given for computing the gradient uncertainty as half the difference between those gradients.
- Marks are given for propagating the gradient uncertainty through any constant multiplier and reporting it with units or significant figures as asked.

Pitfalls

- Joining the points with a dot-to-dot line, or drawing a line that bends to pass through individual points, instead of one straight best-fit line.
- Forcing the line through the origin when the data does not justify it.
- Ignoring the error bars, so the uncertainty comment is not supported by the graph.
- Using the same two points to find the maximum and minimum gradients, or not showing the gradient calculations.
- Forgetting to apply the constant multiplier to the gradient uncertainty, or to give the final answer with appropriate significant figures.

Pattern 17: Communicating with scientific language

Syllabus: S26-SIS-SIS-17 / S27-SIS-SIS-17

Frequency: 0 of 10 exam years; 0 marks in all.

Summary: This point asks you to communicate science to a stated audience and for a stated purpose, using correct terms, units and symbols. No past-paper parts are recorded for it (parts_total 0), so this note is written from the syllabus wording and general physics practice, not from examined questions; it is likely to appear as a clarity or wording mark within longer written answers rather than as a stand-alone question.

Recognise: Look for prompts that name an audience (a peer, a parent, a non-scientist, a technician) or a purpose (explain, justify, report a result, persuade a decision-maker), or that ask you to write a short explanation, summary or report in scientific language.

Steps

1. Identify the audience and purpose first: who is reading, and do they need to understand, decide or reproduce something?
2. Choose the level of detail and vocabulary to suit that audience, defining any technical term a lay reader would not know.
3. Use the correct term for each physical quantity (for example, net force rather than 'push', resistance rather than 'how hard it is to flow') and keep one meaning per word throughout.
4. Write each quantity with its symbol and SI unit, for example speed v in m s^-1, or a relationship such as F = ma or V = IR, stated plainly.
5. Structure the answer so the main claim comes first, followed by the reasoning in a clear chain using connectives such as 'therefore' and 'because'.
6. Check the final text for unit consistency, sign conventions, and whether each term is used the same way in every sentence.

Where the marks are

- A mark for using the correct scientific term or name in place of an everyday word.
- A mark for including units with numerical values and using standard symbols consistently.
- A mark for a clear main statement that answers the purpose set in the question.
- A mark for reasoning that is logically ordered and linked, rather than a list of loosely related facts.
- A mark for pitching the explanation at the stated audience, with any necessary definitions.

Pitfalls

- Using everyday words such as 'power', 'heat' or 'energy' interchangeably with their precise physics meanings.
- Omitting units, or giving units that do not match the quantity (for example, a speed in kg).
- Writing vague comparisons such as 'it gets faster' without a direction, a quantity, or a reason.
- Pitching the answer at the wrong level, either too technical for a lay reader or too simple for a specialist.
- Changing the meaning of a symbol partway through the answer, or using the same symbol for two

different quantities.